

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing customer shopping behavior using transactional data containing **3,900 purchase records** across multiple product categories. The primary objective is to identify meaningful patterns in customer spending, purchasing preferences, customer segments, and subscription behavior.

The analysis aims to generate actionable business insights that can help the organization improve **sales performance, customer engagement, customer retention, and marketing strategies**.

2. Dataset Summary

The dataset consists of **3,900 rows and 18 columns**, covering customer demographics, purchase information, and shopping behavior.

Key Features

- **Customer Demographics:** Age, Gender, Location, and Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, and Color
- **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, and Shipping Type

Missing Data

The dataset contains **37 missing values in the Review Rating column**. These missing values were identified during the data-quality assessment and were considered during the data-cleaning process.

3. Exploratory Data Analysis Using Python

The project initially involved data preparation, quality assessment, and exploratory analysis using **Python and Pandas**.

Data Loading

The dataset was imported into Python using the **Pandas** library, providing a structured environment for data cleaning and analysis.

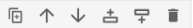
Initial Data Exploration

The structure and quality of the dataset were examined using functions such as:

- `df.info()` — to understand the dataset structure, data types, and non-null values.
- `df.describe()` — to generate descriptive statistics for numerical variables and understand their distributions.

This initial exploration helped identify the key variables, assess data quality, and determine the next steps for data cleaning and analysis.

df.head()



	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Ver
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	(
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Cl
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	Pa
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	Pa

df.describe()

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

df.info()

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 3900 entries, 0 to 3899

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	object
17	Frequency of Purchases	3900 non-null	object

dtypes: float64(1), int64(4), object(13)

memory usage: 548.6+ KB

Data Cleaning and Feature Engineering

- **Missing Data Handling:** Identified missing values in the dataset and handled the **37 missing values in the Review Rating column** by imputing them with the median review rating calculated for each respective product category.
- **Column Standardization:** Renamed the dataset columns using the **snake_case naming convention** to improve readability, consistency, and ease of use during analysis and documentation.
- **Feature Engineering:** Created additional features to support deeper customer behavior analysis:
 - **age_group:** Categorized customers into different age groups based on their age.
 - **purchase_frequency_days:** Derived the purchase frequency in terms of days to enable more detailed analysis of customer purchasing patterns.
- **Data Consistency Check:** Evaluated the relationship between the **discount_applied** and **promo_code_used** columns to identify potential redundancy. Since the information provided by **promo_code_used** was considered redundant for the analysis, the column was removed.
- **Database Integration:** Established a connection between Python and **PostgreSQL** and loaded the cleaned and transformed DataFrame into the database. This enabled further analysis using SQL and provided a structured environment for business-oriented querying.

4. Data Analysis Using SQL — Business Transactions

After completing the data cleaning and database integration process, the cleaned dataset was analyzed using **PostgreSQL SQL queries**.

The objective of this stage was to answer important business questions and identify actionable insights related to customer spending, purchasing patterns, product performance, and customer behavior.

1. Revenue by Gender

Analyzed and compared the **total revenue generated by male and female customers** to understand differences in purchasing contribution across gender segments.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users

Identified customers who **used discounts while making purchases above the overall average purchase amount**. This analysis helps determine whether discounts are successfully encouraging customers to make higher-value purchases and provides insights into the effectiveness of discount-based promotional strategies.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
Total rows: 839		Query complete 00:00:

3. Top 5 Products by Rating

Analyzed the average **review rating for each product** and identified the **top five highest-rated products**. This analysis helps the business understand which products receive the strongest customer feedback and can support decisions related to product promotion, inventory planning, and customer satisfaction.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison

Compared the **average purchase amount across Standard and Express shipping methods** to determine whether customers using different shipping options exhibit differences in spending behavior. This analysis helps the business understand the relationship between **shipping preferences and customer purchase value**, supporting decisions related to shipping services and pricing strategies.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs. Non-Subscribers

Compared the **average purchase amount and total revenue generated by subscribers and non-subscribers** to evaluate differences in spending behavior and overall revenue contribution. This analysis helps determine whether subscription membership is associated with **higher customer spending and revenue generation**, providing insights for improving subscription strategies and customer retention.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Dependent Products

Identified the **top five products with the highest percentage of discounted purchases** to determine which products are most frequently purchased when discounts are applied. This analysis helps the business identify **discount-dependent products** and evaluate opportunities to optimize promotional strategies and pricing decisions.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Segmented customers into **New, Returning, and Loyal** categories based on their **previous purchase history**. This segmentation helps the business understand different levels of customer engagement and identify opportunities for **customer retention, targeted marketing, and loyalty-building strategies**.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category

Identified the **top three most frequently purchased products within each product category** by analyzing purchase frequency across categories. This analysis helps the business understand

product-level demand and customer preferences, supporting inventory planning, product promotion, and category-level sales strategies.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Analyzed whether customers with more than five previous purchases are more likely to have an active subscription. This analysis helps evaluate the relationship between repeat purchasing behavior and subscription adoption, providing insights into customer loyalty and the effectiveness of subscription-based retention strategies.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

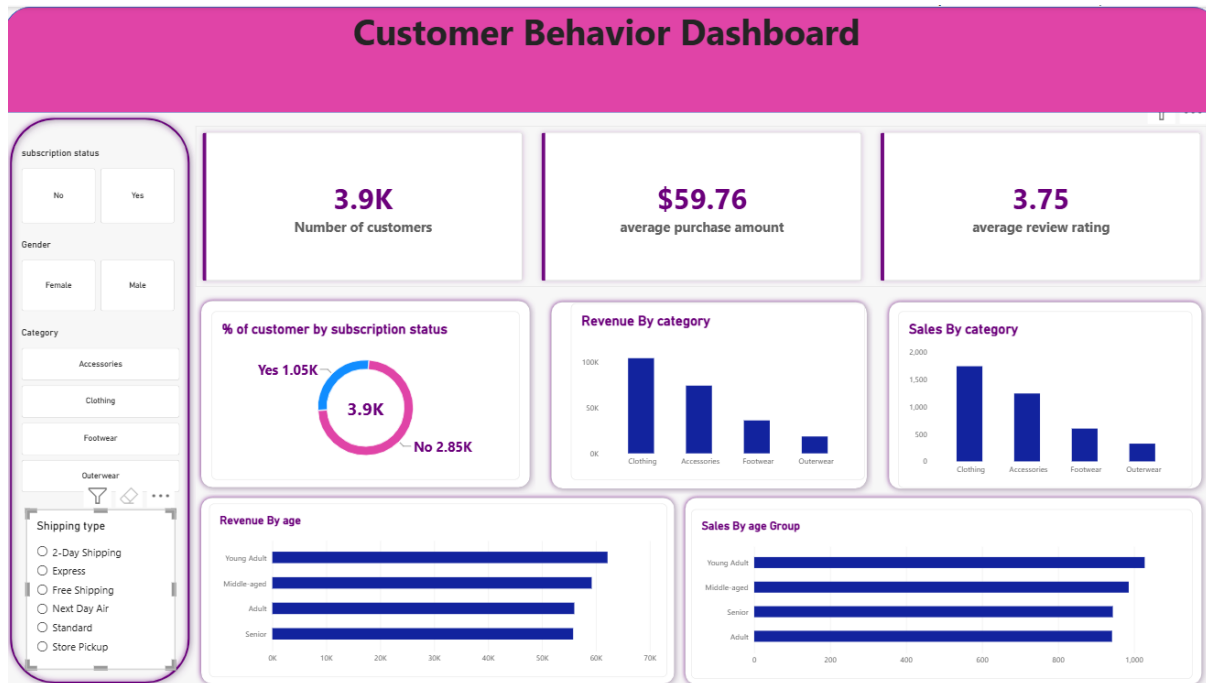
10. Revenue by Age Group

Analyzed total revenue generated by different customer age groups to identify which demographic segments contribute the most to overall sales. This analysis helps the business understand age-based purchasing patterns and develop targeted marketing, product

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Developed an interactive Power BI dashboard to transform the analyzed data into clear and actionable visual insights. The dashboard presents key performance indicators, customer demographics, purchasing patterns, product performance, subscription behavior, and revenue trends through interactive charts, graphs, and filters. This enables business stakeholders to explore the data dynamically and make data-driven decisions.



6. Business Recommendations

- **Increase Subscription Adoption:** Promote exclusive benefits, discounts, and loyalty rewards to encourage more customers to subscribe.
- **Strengthen Customer Loyalty:** Introduce loyalty programs and personalized rewards to convert repeat buyers into loyal customers.
- **Optimize Discount Strategy:** Evaluate discount effectiveness by product and customer segment while maintaining healthy profit margins.
- **Promote High-Performing Products:** Highlight top-rated and best-selling products in marketing campaigns and promotional activities.
- **Target High-Value Age Groups:** Focus marketing campaigns and product offers on age groups generating higher revenue and sales.
- **Optimize Shipping Strategy:** Analyze customer preferences for shipping options and promote suitable express-shipping services where they can improve customer experience and order value.